

WEEK 3 | OPTIONAL PRACTICE MODULE

HEXSTRIKE MCP WITH CLAUDE DESKTOP

Project information

- Name: Rachel Debbarma
- Date: 27 August 2026

1. Liability Disclaimer

This report documents an authorized cybersecurity training exercise completed as part of the Networkwalks Cybersecurity & Ethical Hacking internship. The tools and techniques were used within the provided learning environment and against a practice PDF. The activities described are intended for education, security testing, and authorized use only.

2. Introduction

This report covers the optional Week 3 practice module on setting up HexStrike MCP with Claude Desktop on a Kali Linux virtual machine. The module introduced an AI-assisted workflow in which Claude Desktop communicates with cybersecurity tools through the HexStrike-AI MCP server.

After setting up and connecting HexStrike to Claude, I used the environment to interact with John the Ripper during a controlled PDF password-recovery exercise. This provided practical experience with MCP configuration, local cybersecurity tooling, Linux troubleshooting, and AI-assisted security workflows.

3. Objective

- Set up Claude Desktop in Kali Linux.
- Install and configure the HexStrike-AI MCP server.
- Create a Python virtual environment for HexStrike.
- Connect HexStrike-AI with Claude Desktop through MCP.
- Verify that the HexStrike server and its tools were operational.
- Verify the John the Ripper installation.
- Use the HexStrike/JTR workflow on the authorized practice PDF.
- Understand how an AI assistant can interact with local cybersecurity tools.

4. Lab Environment

Component	Details
Operating System	Kali Linux
AI Assistant	Claude Desktop
MCP Server	HexStrike-AI
MCP Endpoint	http://localhost:8888
Password Tool	John the Ripper
Wordlist	rockyou.txt
Target used	My Locked PDF1.pdf
Target size	66 KB

5. Tools Used

- Kali Linux
- Claude Desktop
- HexStrike-AI MCP Server
- Python virtual environment
- John the Ripper (JTR)
- RockYou wordlist
- Linux terminal utilities

6. Activities Performed

6.1 Installing Claude Desktop

Claude Desktop was installed and opened inside the Kali Linux environment. The application was then used as the interface for the HexStrike MCP integration. The setup guide specifies Claude Desktop as part of the required lab environment. [filecite turn6file15](#)

6.2 Downloading and Setting Up HexStrike-AI

HexStrike-AI was cloned from its GitHub repository and a dedicated Python virtual environment was created.

```
git clone https://github.com/0x4m4/hexstrike-ai.git
cd hexstrike-ai
python3 -m venv hexstrike-env
source hexstrike-env/bin/activate
```

6.3 Installing Dependencies and Troubleshooting

The required dependencies were installed using the project's requirements file. The initial installation failed with a 'No space left on device' error while building angr/pyvex dependencies.

ERROR: [Errno 28] No space left on device

The main filesystem still had free space, while /tmp was mounted as a small temporary filesystem. I created a temporary build directory on the main filesystem and directed package builds there.

```
mkdir -p ~/pip-tmp
export TMPDIR=$HOME/pip-tmp
```

After this change, the dependencies installed successfully. The earlier incomplete installation had also caused a Flask ModuleNotFoundError when the server was started; completing the dependency installation resolved it.

6.4 Starting the HexStrike Server

```
cd ~/hexstrike-ai
source hexstrike-env/bin/activate
python3 hexstrike_server.py
```

The HexStrike server was successfully started on the local Kali environment.

6.5 Configuring Claude Desktop

Claude Desktop was configured with the HexStrike MCP server. The configuration used the Python interpreter from the HexStrike virtual environment and the local server endpoint.

```
{
  "mcpServers": {
    "hexstrike-ai": {
      "command": "/home/kali/hexstrike-ai/hexstrike-env/bin/python",
      "args": [
        "/home/kali/hexstrike-ai/hexstrike_mcp.py",
        "--server",
        "http://localhost:8888"
      ]
    }
  }
}
```

6.6 Verifying the HexStrike MCP Server

A health-check request was made through Claude Desktop. The result confirmed that the HexStrike AI MCP server was healthy and operational.

Metric	Observed Result
Status	Healthy and operational
HexStrike version	6.0.0
Essential tools	8/8 available
Tools available	60/127 total
Password category	4/5 available

6.7 Verifying John the Ripper

John the Ripper was checked from the Kali terminal and was found to be installed and executable.

John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c

6.8 Preparing the Practice PDF

For the actual exercise, I used the local file '/home/kali/Desktop/My Locked PDF1.pdf'. The file was verified as present and readable, with a size of approximately 66 KB.

```
ls -lh "/home/kali/Desktop/My Locked PDF1.pdf"
```

6.9 AI-Assisted JTR Workflow

After the MCP connection was working, Claude was instructed to use the JTR functionality available through HexStrike for the controlled PDF password-recovery task. The PDF hash was extracted and John the Ripper was run using the RockYou wordlist. The optional Networkwalks AI lab describes this as a JTR + AI workflow using HexStrike-AI MCP with Claude Desktop on Kali Linux.

The successful response reported that the hash had been extracted and that John the Ripper was running against it with rockyou.txt. The password was then recovered successfully.

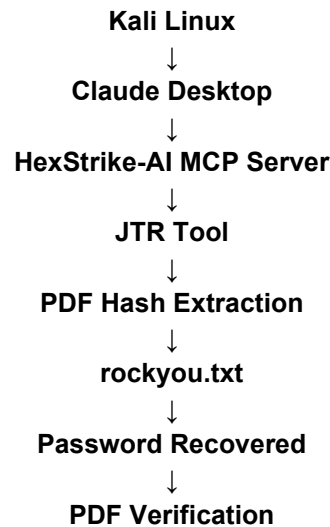
6.10 Password Recovery and Verification

The recovered password was:

password1

The password was used to open the protected PDF, confirming that the recovery process was successful.

7. Overall Workflow



8. Challenges Faced

- Dependency installation initially failed with a 'No space left on device' error while building angr/pyvex.
- The main filesystem had sufficient space, but /tmp was a small tmpfs; setting TMPDIR to a directory on the main filesystem allowed the build to complete.
- The incomplete dependency installation initially caused a Flask ModuleNotFoundError when starting HexStrike.
- Claude initially did not display the MCP server because the MCP configuration needed to be placed in the correct Claude Desktop configuration file.
- Restarting Claude Desktop after correcting the configuration allowed HexStrike to load successfully.

9. Results

The optional HexStrike practice module was completed successfully. Claude Desktop was connected to the local HexStrike-AI MCP server, the server health check reported a healthy operational state, John the Ripper was available, and the controlled PDF password-recovery task completed successfully.

Task	Result
Claude Desktop setup	Completed
HexStrike-AI setup	Completed
MCP configuration	Completed
MCP health check	Successful
JTR verification	Successful
PDF hash extraction	Successful
JTR + RockYou workflow	Successful
Password recovered	password1
PDF verification	Successful

10. Risk Analysis / Impact

Finding	Observation	Potential Impact	Risk
Weak password	The lab password was recovered through a dictionary-based	Common passwords can be identified more easily.	Medium

	workflow.		
Password-cracking exposure	A PDF hash can be used for password testing when the password is weak.	Weak document passwords may provide limited protection against offline testing.	Medium
Tool-assisted automation	MCP allowed an AI assistant to interact with local security tooling.	Automation can speed up security workflows, making authorization and scope controls important.	Medium

11. Recommendations

- Use long, unique, and unpredictable passwords for protected documents.
- Avoid common passwords and predictable patterns such as 'password1'.
- Use a password manager to generate and store strong passwords.
- Restrict password-cracking activities to authorized security assessments and lab environments.
- Keep MCP servers and security tooling isolated from systems outside the assessment scope.
- Document configuration changes and troubleshooting steps when building cybersecurity lab environments.

12. What I Learned

- How MCP can connect an AI assistant with locally available cybersecurity tools.
- How to set up and configure HexStrike-AI on Kali Linux.
- How Claude Desktop can load and communicate with an MCP server.
- How to verify an MCP server using a health check.
- How John the Ripper can be accessed as part of an AI-assisted workflow.
- How Linux temporary storage can affect Python package installation and how to troubleshoot it.
- The importance of correct configuration files and restarting applications after configuration changes.
- Why weak passwords remain vulnerable to dictionary-based attacks.

13. Evidence Collected

- HexStrike server running on Kali Linux.
- Claude Desktop MCP configuration.
- HexStrike MCP health-check response showing healthy and operational status.
- JTR version output.
- Target PDF and file details.
- Claude response showing successful hash extraction and JTR execution.
- Final response showing the recovered password.
- Protected PDF opened successfully using the recovered password.

14. Conclusion

The optional Week 3 HexStrike module provided practical experience with integrating an AI assistant with cybersecurity tools through the Model Context Protocol. I successfully set up HexStrike-AI on Kali Linux, connected it to Claude Desktop, resolved dependency and configuration issues, and verified that the MCP server was operational.

I then used the working HexStrike environment to interact with John the Ripper during a controlled PDF password-recovery exercise. The password was successfully recovered as 'password1' and verified by opening the protected PDF. Overall, the exercise helped me understand how MCP can connect AI systems with security tools while giving me practical experience in Linux troubleshooting, tool configuration, and responsible security testing.

15. References

Networkwalks Academy – How to setup HexStrike MCP Server (Practice Lab).

Networkwalks Academy – JTR Password Cracking Lab v1 (AI-version).

HexStrike-AI project repository: <https://github.com/0x4m4/hexstrike-ai>

Project repository: <https://github.com/rach-db/networkwalks-B082-week3>

End of Report